



Probability Theory and Statistical Inference

Axiomatic Foundations, Limit Theorems, Density Modeling, and Association Tests

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Statistical Inquiry Roadmap

CURRICULUM

- **1. Probability Axioms:** Conditional spaces and Bayes' rule [1].
- **2. Random Moments:** Expectations, variance, and covariances.
- **3. Asymptotic Limit Theorems:** LLN and Central Limit Theorem.
- **4. Density Modeling:** Gaussian distribution and empirical rule.
- **5. Categorical Association:** Contingency tables and χ^2 tests [2].

Core Objective: Formulate exact probability models and derive sample estimators with known asymptotic distributions.

SAMPLE SIZE

$N = 1,200$

Cohort Total

CONFIDENCE

95.0%

Coverage Interval

Probability Axioms & Bayesian Inversion

AXIOMATIC SPACE

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

- **Non-negativity:** $P(A) \geq 0 \quad \forall A \in \mathcal{F}$.
- **Total Mass:** $P(\Omega) = 1$.
- **Countable Additivity:** $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ for disjoint A_i .

Conditional Probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

BAYES' THEOREM

Let $\{B_1, \dots, B_k\}$ partition Ω :

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^k P(A | B_j)P(B_j)}$$

- **Prior** $P(B_i)$: Baseline belief before observation.
- **Likelihood** $P(A | B_i)$: Evidence generation probability.
- **Posterior** $P(B_i | A)$: Updated belief given evidence A .

Key Takeaway: Bayes' rule inverts conditional dependencies to infer latent parameters from observable data.

Random Moments & The Central Limit Theorem

MOMENT OPERATORS

1. Expected Value (First Moment):

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x) dx$$

2. Variance (Second Central Moment):

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

3. Covariance Matrix:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

LIMIT CONVERGENCE

Lindeberg-Lévy Central Limit Theorem: For i.i.d. $X_1, \dots, X_n$ with mean μ and variance $\sigma^2 < \infty$:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty$$

- **Weak Law (WLLN):** $\bar{X}_n \xrightarrow{P} \mu$.
- **Rate of Convergence:** Sampling error shrinks at parametric rate $\mathcal{O}(n^{-1/2})$.

Standard Normal Distribution & Hypothesis Rejection

DENSITY MODEL

Standard Normal Density:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

■ Empirical Rule (68-95-99.7):

- $P(|Z| \leq 1) = 68.27\%$
- $P(|Z| \leq 2) = 95.45\%$
- $P(|Z| \leq 3) = 99.73\%$

■ Type I Error Threshold ($\alpha = 0.05$): Two-tailed critical values $z_{\text{crit}} = \pm 1.96$.

■ Two-Tailed p -Value: $p = 2 \cdot [1 - \Phi(|z_{\text{stat}}|)]$.

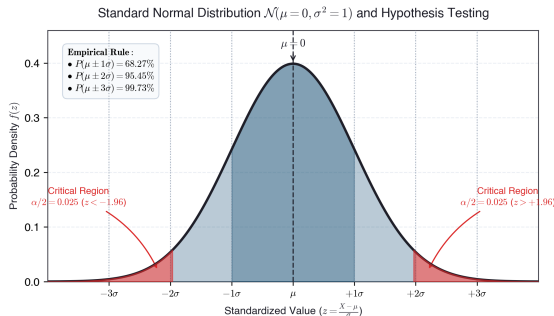


Fig 1: High-DPI Python/Seaborn Standard Normal simulation with rejection regions.

Contingency Matrices & Pearson's χ^2 Test

Cohort Group	Outcome ($Y = 1$)	Control ($Y = 0$)	Total (R_i)
Treatment ($D = 1$)	$n_{11} = 420$	$n_{12} = 180$	600
Control ($D = 0$)	$n_{21} = 210$	$n_{22} = 390$	600
Total (C_j)	630	570	$N = 1,200$

Cross-tabulation of sample cohorts with marginal frequencies.

Decision: $\chi^2_{\text{calc}} = 147.37 \gg \chi^2_{0.05,1} = 3.84 \implies \text{Reject } H_0$
($p < 0.001$).

TEST FORMULATION

- Expected Frequency (E_{ij}):

$$E_{ij} = \frac{R_i \times C_j}{N}$$

- Pearson's χ^2 Statistic:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)}$$

- Sample Odds Ratio ($\widehat{\text{OR}}$):

$$\widehat{\text{OR}} = \frac{n_{11}n_{22}}{n_{12}n_{21}} = \frac{420 \times 390}{180 \times 210} = 4.33$$

Statistical Estimation in Python

```

1 import numpy as np
2 from scipy import stats
3
4 def perform_contingency_analysis(table: np.ndarray):
5     """Computes Chi-Square statistic, p-value, and Odds Ratio."""
6     chi2, p_val, dof, expected = stats.chi2_contingency(table, correction=False)
7
8     # 2x2 Odds Ratio calculation
9     a, b = table[0, 0], table[0, 1]
10    c, d = table[1, 0], table[1, 1]
11    odds_ratio = (a * d) / (b * c)
12    se_ln_or = np.sqrt(1/a + 1/b + 1/c + 1/d)
13    ci_95 = np.exp(np.log(odds_ratio) + np.array([-1, 1]) * 1.96 * se_ln_or)
14
15    return {"chi2": chi2, "p_value": p_val, "odds_ratio": odds_ratio, "ci_95": ci_95}

```

Key Statistical Takeaways

SUMMARY

- **Axiomatic Rigor:** Invert conditional probabilities via Bayes' rule.
- **CLT Asymptotics:** Normality emerges regardless of underlying distribution.
- **Categorical Testing:** χ^2 tests evaluate independence against random marginal allocation.

NEXT STEPS

- **Problem Set 1:** Derive the CLT for exponential distributions.
- **Lab 1:** Simulate empirical coverage of $\widehat{OR}$ confidence intervals.
- **Office Hours:** Tuesday 10:00–12:00 (Math Hall 302).

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References i

- [1] William Feller. *An Introduction to Probability Theory and Its Applications*. Vol. 1. John Wiley & Sons, 1968.
- [2] George Casella and Roger L. Berger. *Statistical Inference*. Duxbury Press, 2002.